reportVersion: satmod-analysis-v2
id: 1049c028-4cbd-454e-b0e8-07913f8f3149
createdAt: 2026-11-12T14:44:49.507633+00:00
source: Generated report fixture, not real RF
captureId: 7f8fc310-4d6c-434b-8a66-621b73fdfaa1
capturePosition: {"captureBoundariesWithinWindow": [], "captureBoundaryNote": null, "centerFrequency": null, "fileSample": 0, "globalSampleIndex": 0, "globalSampleIndexSource": "Recording sample index; no independent hardware counter supplied", "seconds": 0.0, "sourceSample": 0, "timestamp": null, "windowCrossesCaptureSegment": false}
inputHash: 1fb21c61da6eeb8f2e712ad667c0a857eaea9258a379bb0a579c749f15b566ff
inputRetention: Managed recording reference
sampleRate: 8000
modelId: 1d-cnn
modelVersion: supplied-cnn-v1
modelHash: e9705c98275e26d4a60210b2d563c4cc32ef94f4f0368974f1168245f91cf33d
preprocessing: {"amplitudePolicy": "Preserve samples; training z-score only", "mean": -6.560127076227218e-05, "std": 0.006070038769394159, "windowLength": 128, "windowStart": 0}
predictedModulation: QPSK
confidence: 0.5449240207672119
entropyBits: 1.3228542016218974
processingMs: 3.907649021130055
probabilities: {"16-QAM": 0.09109178930521011, "BPSK": 0.3639841377735138, "QPSK": 0.5449240207672119}
quantityTypes: {"metrics": "computed from samples", "probabilities": "uncalibrated softmax", "snr": "unavailable unless supplied independently"}
Confidence and class scores are raw, uncalibrated softmax outputs; they are not calibrated correctness probabilities.

CAPTURE PROVENANCE

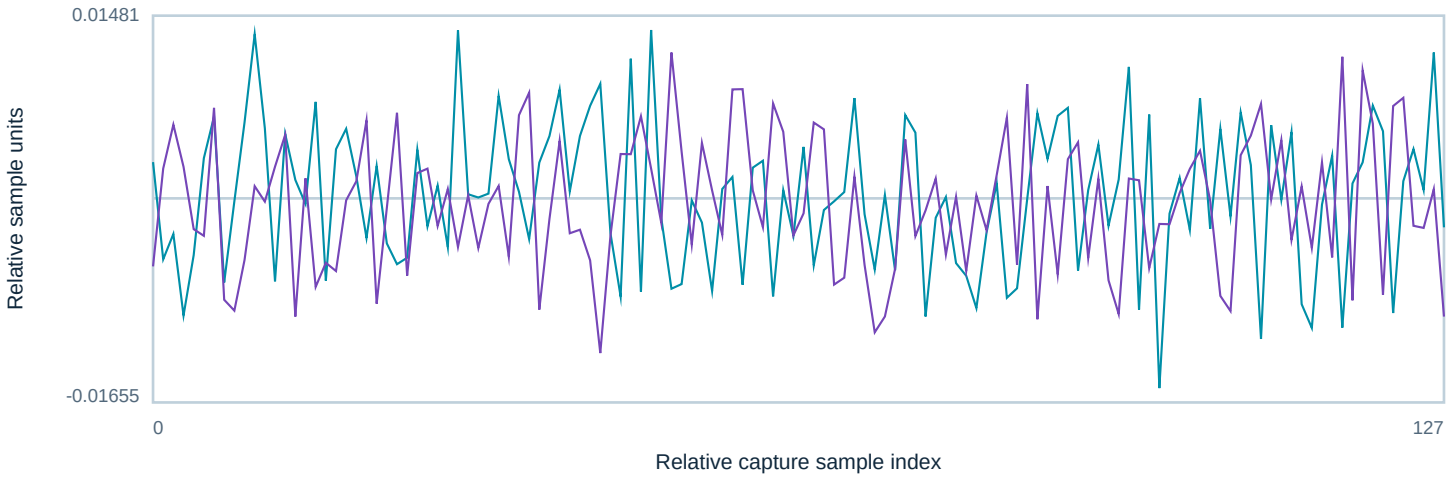
id: 7f8fc310-4d6c-434b-8a66-621b73fdfaa1
source: Generated report fixture, not real satellite RF
sessionId: synthetic-report-example
hash: 23e92ab698b4ec11542212c8820702e6a0a3bc71373846754efb1717cfe560e2
sourceHash: 8b855a0d25019326f9ce0a3446ec9630d98f765bf343b42b4835b0a2f826de0c
sampleCount: 512
sampleRate: 8000.0
sampleRateKnown: True
sourceOffset: 0
centerFrequency: None
datatype: parsed application format
layout: interleaved
iqOrder: IQ
scale: 1
license: Synthetic demonstration fixture

COMPUTED SIGNAL METRICS

sampleCount: 128
sampleRate: 8000
signalPower: 6.51925582297108e-05
powerDb: -41.85801976418816

rmsAmplitude: 0.008074190871518384
bandwidthHz: 7937.5
noisePower: None
estimatedSnrDb: None
qualityNote: Noise and SNR cannot be uniquely inferred from an arbitrary capture without a reference or noise-only region. Power is in relative sample units (not calibrated watts).
I_Mean: 0.00020690464342010273
Q_Mean: -0.00041398694474992226
I_Std: 0.005923163187133505
Q_Std: 0.005467586429715135
Amplitude_Mean: 0.007345773624582725
Amplitude_Std: 0.00335144268727597
Phase_Std_rad: 1.754891849590334
Kurtosis_I: -0.466679028429116
Kurtosis_Q: -0.7507111882594977
Skewness_I: -0.021484647613806623
Skewness_Q: 0.0368309780460318
Signal_Power_dB_relative: -41.85801976418816
I_Variance: 3.508386214141354e-05
Q_Variance: 2.9894501366405093e-05
Crest_Factor: 1.933115085935364
PAPR_dB: 5.725154201097244

SELECTED I/Q WAVEFORM

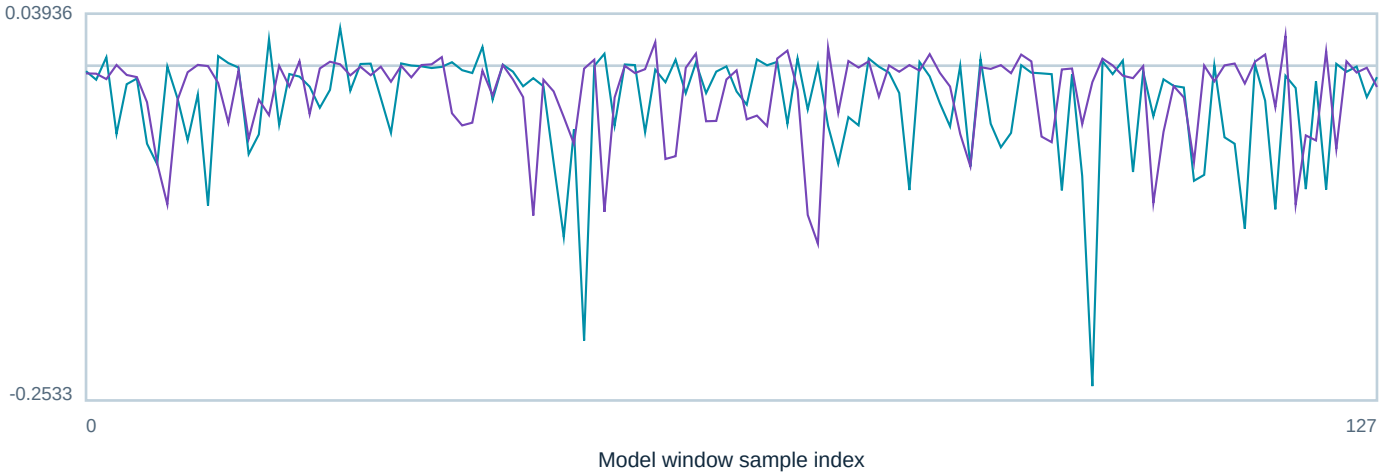


I | Q

INTEGRATED-GRADIENTS / QPSK

createdAt: 2026-11-12T14:44:49.640125+00:00
method: integrated-gradients
targetClass: QPSK
targetQuantity: pre-softmax class logit
modelId: 1d-cnn
modelVersion: supplied-cnn-v1
modelHash: e9705c98275e26d4a60210b2d563c4cc32ef94f4f0368974f1168245f91cf33d
inputHash: 1fb21c61da6eeb8f2e712ad667c0a857eaea9258a379bb0a579c749f15b566ff
windowStart: 0
windowLength: 128
integrationSteps: 32
baseline: {"description": "Zero raw I/Q capture transformed by the same training z-score", "normalizedValue": 0.010807389088359931, "rawValue": 0.0}
inputLogit: 1.1720194816589355
baselineLogit: 7.243814468383789
logitDifference: -6.0717949867248535
attributionSum: -6.13962514782708
completenessError: -0.0678301611022265
processingMs: 92.05185697646812
quantityUnit: class-logit contribution
Signed additive class-logit contributions relative to the recorded baseline.

SIGNED I/Q CLASS-LOGIT ATTRIBUTION



I | Q

SIGNED CHANNEL SUMMARIES

I: {"absoluteSum": 3.798560772657556, "negativeSum": -3.657598558899552, "positiveSum": 0.14096221375800413}
Q: {"absoluteSum": 2.949252907370025, "negativeSum": -2.7861208550277787, "positiveSum": 0.16313205234224631}

MODEL-RANKED SAMPLE REGIONS

At most 16 highest-absolute-importance samples, joined only when contiguous. These are model-ranked regions, not detected RF packets or statistical significance.
Samples [7, 8) | absolute importance 0.148423 | share 2.20%

Samples [16, 17) | absolute importance 0.12207 | share 1.81%
Samples [44, 45) | absolute importance 0.122839 | share 1.82%
Samples [47, 48) | absolute importance 0.168483 | share 2.50%
Samples [49, 50) | absolute importance 0.210195 | share 3.12%
Samples [71, 73) | absolute importance 0.281 | share 4.16%
Samples [87, 88) | absolute importance 0.152439 | share 2.26%
Samples [98, 100) | absolute importance 0.382606 | share 5.67%
Samples [105, 106) | absolute importance 0.14216 | share 2.11%
Samples [109, 110) | absolute importance 0.160033 | share 2.37%
Samples [114, 115) | absolute importance 0.136845 | share 2.03%
Samples [117, 118) | absolute importance 0.140043 | share 2.08%
Samples [119, 121) | absolute importance 0.268187 | share 3.97%

Limitation: Attributions describe this model relative to the chosen baseline; they do not establish physical causality or correctness.

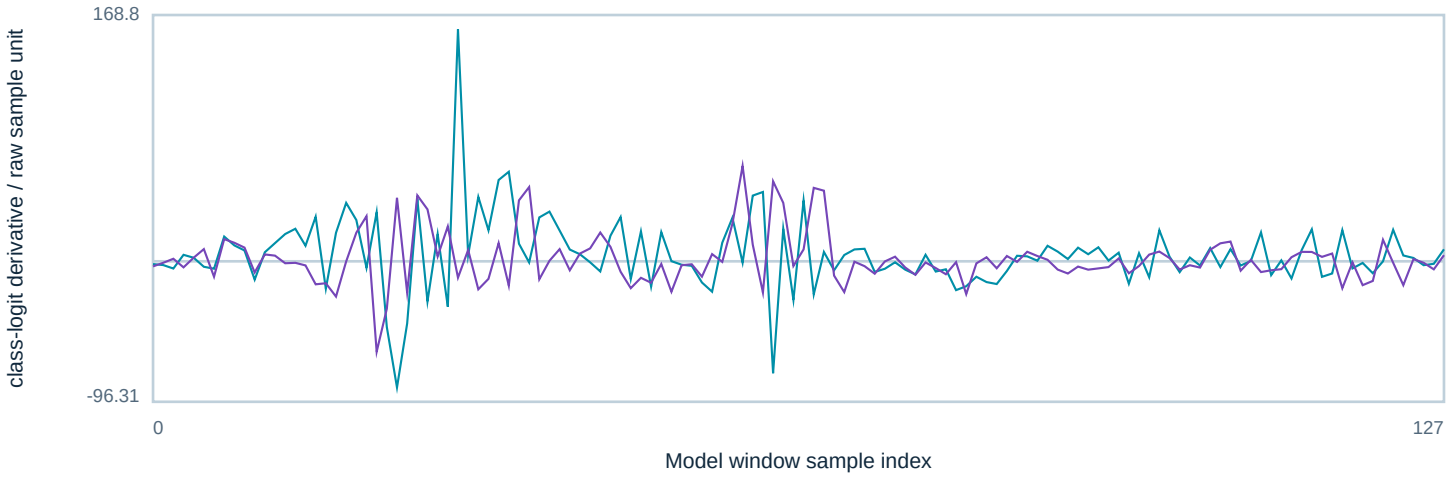
Limitation: Integrated Gradients uses finite trapezoidal integration; inspect the reported completeness residual.

Limitation: Saliency is a local derivative in raw sample units; it is not an additive contribution map.

SALIENCY / QPSK

createdAt: 2026-11-12T14:44:49.644383+00:00
method: saliency
targetClass: QPSK
targetQuantity: pre-softmax class logit
modelId: 1d-cnn
modelVersion: supplied-cnn-v1
modelHash: e9705c98275e26d4a60210b2d563c4cc32ef94f4f0368974f1168245f91cf33d
inputHash: 1fb21c61da6eeb8f2e712ad667c0a857eaea9258a379bb0a579c749f15b566ff
windowStart: 0
windowLength: 128
integrationSteps: 0
baseline: {"description": "Zero raw I/Q capture transformed by the same training z-score", "normalizedValue": 0.010807389088359931, "rawValue": 0.0}
inputLogit: 1.1720194816589355
baselineLogit: 7.243814468383789
logitDifference: -6.0717949867248535
attributionSum: None
completenessError: None
processingMs: 3.5183410509489477
quantityUnit: class-logit derivative / raw sample unit
Signed local class-logit derivatives per raw sample unit; channel sums are derivative summaries, not additive contributions.

SIGNED I/Q LOCAL LOGIT DERIVATIVES



SIGNED CHANNEL SUMMARIES

I: {"absoluteSum": 2040.6383768729866, "negativeSum": -724.5646815188229, "positiveSum": 1316.0736953541636}
Q: {"absoluteSum": 1501.046891219914, "negativeSum": -649.9292540922761, "positiveSum": 851.1176371276379}

MODEL-RANKED SAMPLE REGIONS

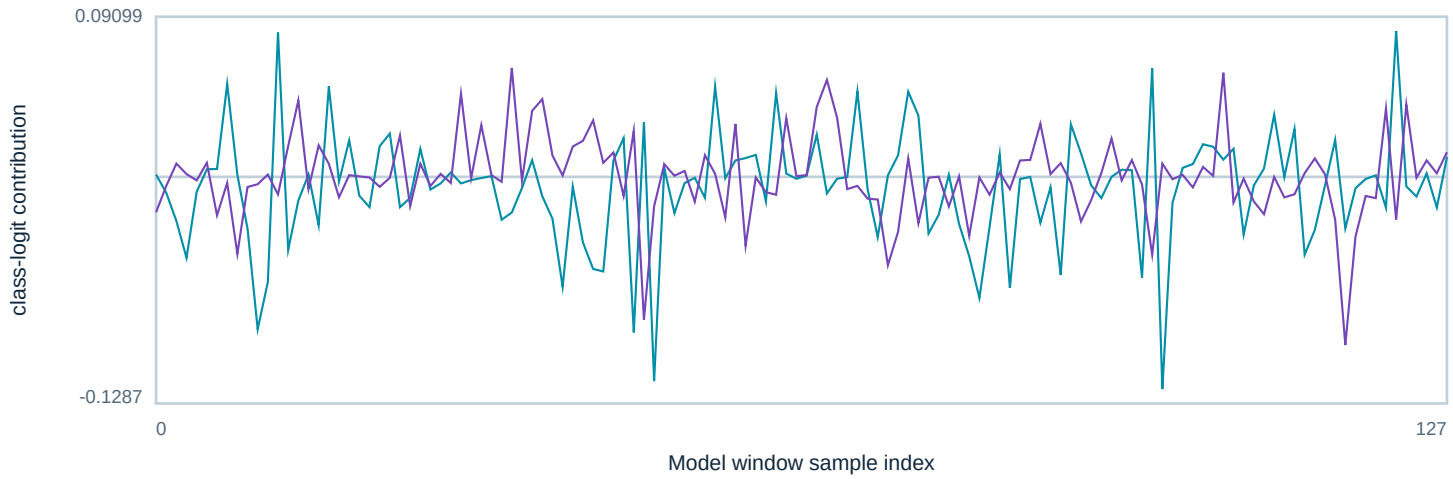
At most 16 highest-absolute-importance samples, joined only when contiguous. These are model-ranked regions, not detected RF packets or statistical significance.

Samples [22, 28) | absolute importance 518.217 | share 14.63%
Samples [30, 31) | absolute importance 170.211 | share 4.81%
Samples [32, 33) | absolute importance 63.3518 | share 1.79%
Samples [34, 36) | absolute importance 146.266 | share 4.13%
Samples [58, 63) | absolute importance 385.504 | share 10.88%
Samples [65, 66) | absolute importance 72.98 | share 2.06%
Limitation: Attributions describe this model relative to the chosen baseline; they do not establish physical causality or correctness.
Limitation: Integrated Gradients uses finite trapezoidal integration; inspect the reported completeness residual.
Limitation: Saliency is a local derivative in raw sample units; it is not an additive contribution map.

INTEGRATED-GRADIENTS / BPSK

createdAt: 2026-11-12T14:44:50.133902+00:00
method: integrated-gradients
targetClass: BPSK
targetQuantity: pre-softmax class logit
modelId: 1d-cnn
modelVersion: supplied-cnn-v1
modelHash: e9705c98275e26d4a60210b2d563c4cc32ef94f4f0368974f1168245f91cf33d
inputHash: 1fb21c61da6eeb8f2e712ad667c0a857eaea9258a379bb0a579c749f15b566ff
windowStart: 0
windowLength: 128
integrationSteps: 128
baseline: {"description": "Zero raw I/Q capture transformed by the same training z-score",
"normalizedValue": 0.010807389088359931, "rawValue": 0.0}
inputLogit: 0.7684833407402039
baselineLogit: 1.5104221105575562
logitDifference: -0.7419387698173523
attributionSum: -0.7292604010320415
completenessError: 0.012678368785310812
processingMs: 452.7905319700949
quantityUnit: class-logit contribution
Signed additive class-logit contributions relative to the recorded baseline.

SIGNED I/Q CLASS-LOGIT ATTRIBUTION



I | Q

SIGNED CHANNEL SUMMARIES

I: {"absoluteSum": 2.863028781604772, "negativeSum": -1.8182358744854423, "positiveSum": 1.0447929071193296}
Q: {"absoluteSum": 1.9704029763061952, "negativeSum": -0.9631102049860618, "positiveSum": 1.0072927713201334}

MODEL-RANKED SAMPLE REGIONS

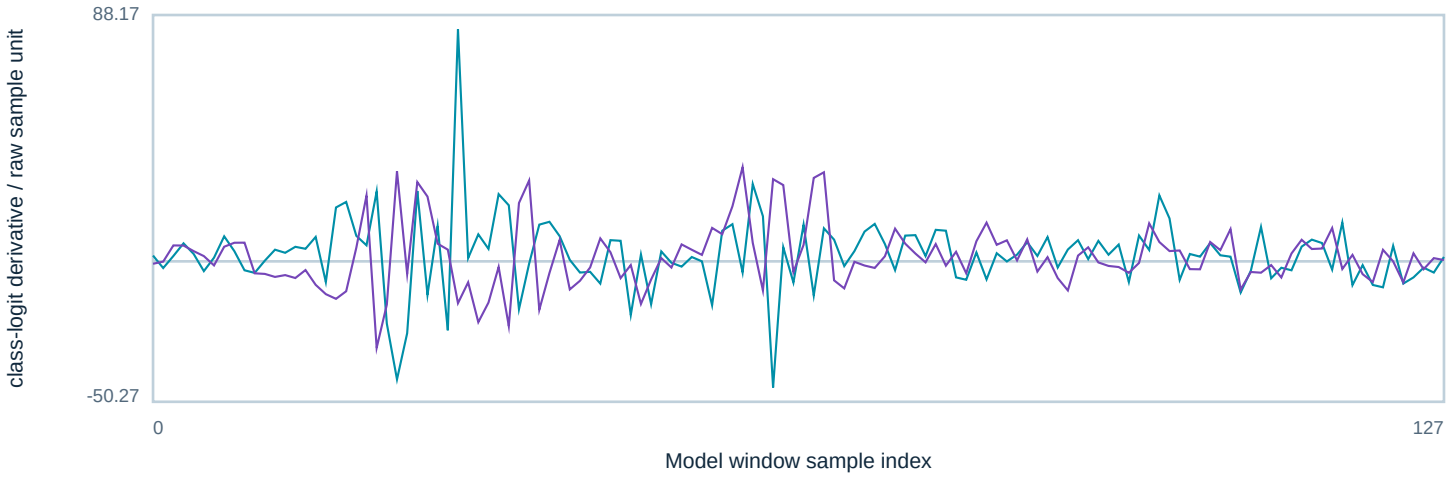
At most 16 highest-absolute-importance samples, joined only when contiguous. These are model-ranked regions, not detected RF packets or statistical significance.
Samples [10, 11) | absolute importance 0.0907235 | share 1.88%

Samples [12, 13) | absolute importance 0.0919362 | share 1.90%
Samples [35, 36) | absolute importance 0.0821231 | share 1.70%
Samples [43, 44) | absolute importance 0.0844017 | share 1.75%
Samples [47, 50) | absolute importance 0.359229 | share 7.43%
Samples [66, 67) | absolute importance 0.0644457 | share 1.33%
Samples [80, 82) | absolute importance 0.147691 | share 3.06%
Samples [84, 85) | absolute importance 0.0700902 | share 1.45%
Samples [98, 100) | absolute importance 0.233913 | share 4.84%
Samples [105, 106) | absolute importance 0.0689263 | share 1.43%
Samples [117, 118) | absolute importance 0.1248 | share 2.58%
Samples [122, 123) | absolute importance 0.107045 | share 2.21%
Limitation: Attributions describe this model relative to the chosen baseline; they do not establish physical causality or correctness.
Limitation: Integrated Gradients uses finite trapezoidal integration; inspect the reported completeness residual.
Limitation: Saliency is a local derivative in raw sample units; it is not an additive contribution map.

SALIENCY / BPSK

createdAt: 2026-11-12T14:44:50.138491+00:00
method: saliency
targetClass: BPSK
targetQuantity: pre-softmax class logit
modelId: 1d-cnn
modelVersion: supplied-cnn-v1
modelHash: e9705c98275e26d4a60210b2d563c4cc32ef94f4f0368974f1168245f91cf33d
inputHash: 1fb21c61da6eeb8f2e712ad667c0a857eaea9258a379bb0a579c749f15b566ff
windowStart: 0
windowLength: 128
integrationSteps: 0
baseline: {"description": "Zero raw I/Q capture transformed by the same training z-score", "normalizedValue": 0.010807389088359931, "rawValue": 0.0}
inputLogit: 0.7684833407402039
baselineLogit: 1.5104221105575562
logitDifference: -0.7419387698173523
attributionSum: None
completenessError: None
processingMs: 3.887327970005572
quantityUnit: class-logit derivative / raw sample unit
Signed local class-logit derivatives per raw sample unit; channel sums are derivative summaries, not additive contributions.

SIGNED I/Q LOCAL LOGIT DERIVATIVES



I | Q

SIGNED CHANNEL SUMMARIES

I: {"absoluteSum": 1148.9512306882534, "negativeSum": -419.14491783827543, "positiveSum": 729.806312849978}
Q: {"absoluteSum": 1030.982998618856, "negativeSum": -407.5044458899647, "positiveSum": 623.4785527288914}

MODEL-RANKED SAMPLE REGIONS

At most 16 highest-absolute-importance samples, joined only when contiguous. These are model-ranked regions, not detected RF packets or statistical significance.

Samples [18, 19) | absolute importance 32.7065 | share 1.50%
Samples [22, 25) | absolute importance 168.228 | share 7.72%
Samples [26, 28) | absolute importance 88.7254 | share 4.07%
Samples [30, 31) | absolute importance 98.0114 | share 4.50%
Samples [35, 37) | absolute importance 81.2657 | share 3.73%
Samples [57, 60) | absolute importance 104.949 | share 4.81%
Samples [61, 63) | absolute importance 106.899 | share 4.90%
Samples [65, 67) | absolute importance 85.774 | share 3.93%
Limitation: Attributions describe this model relative to the chosen baseline; they do not establish physical causality or correctness.
Limitation: Integrated Gradients uses finite trapezoidal integration; inspect the reported completeness residual.
Limitation: Saliency is a local derivative in raw sample units; it is not an additive contribution map.

METHOD RANK COMPARISON

status: available
targetClass: QPSK
measure: Spearman correlation of absolute per-sample importance rankings
spearmanRho: 0.11279985350668374
note: Descriptive ranking comparison only. Raw derivatives and integrated contributions have different units; their magnitudes are not normalized into equivalent quantities. Constant rankings have no defined correlation.
status: available
targetClass: BPSK
measure: Spearman correlation of absolute per-sample importance rankings
spearmanRho: 0.16586362387841053
note: Descriptive ranking comparison only. Raw derivatives and integrated contributions have different units; their magnitudes are not normalized into equivalent quantities. Constant rankings have no defined correlation.
Machine-readable JSON retains all 128 signed I/Q values, raw input values and diagnostics for every stored explanation.

SCIENTIFIC LIMITATIONS

Closed-set three-class classifier; confidence is uncalibrated.
Attribution does not prove physical causality.
No SNR is inferred from arbitrary samples. No validated production OOD decision or physical-causality claim is made.